THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING: LEVEL I EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL1105: Environmental Engineering

DEC 17, 2025 TIME: 9:30 am - 12:30 pm

Answer ALL questions. Section A carries 30 marks and Section B carries 70 marks.

Use of Calculators:

Only approved calculators, as announced by the Examinations Secretary, can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Section A. (Ten single-choice questions with 3 marks each. There will not be any deduction for wrong answers.)
You may need to use Appendix A at the end of the exam paper for your calculations.

1. CO in the atmosphere has reached a record high of 427 ppm this year. What is the CO concentration in the unit of g/m^3 at standard pressure and room temperature?
(A) 0.43 g/m^3; (B) 2.24 g/m^3; (C) 0.25 g/m^3; (D) 0.77 g/m^3.

2. What will be the correct change of species in a close carbonate system if pH decreases from 10 to 6 at standard pressure and room temperature?
(A) $[H_2CO_3]$ increases; (B) $[HCO_3^-]$ increases; (C) $[CO_3^{2-}]$ increases; (D) $[OH^-]$ increases.

3. If 400 kJ of energy is input to heat 1 L of 10 °C cold water, what is the final temperature of the water?
(A) 96 °C; (B) 106 °C; (C) 100 °C; (D) 11 °C.

4. Which of the following statements regarding water is **True**?
(A) The smallest unit that retains the chemical properties of water is the hydrogen and oxygen atoms;
(B) Groundwater occupies around 30.1% of the Earth's water;
(C) Microplastics are a kind of water pollutant as they contaminate water, harming aquatic life;
(D) Pathogens could be one of the water pollutants, and their source is mainly from leaded gasoline.

5. By volume, the concentration of oxygen in the air is about 21%. Find the saturated DO value concentration (i.e., DO_s) in freshwater at 1 atmosphere of pressure.
(A) 2.65×10^{-4} mol; (B) 4.25 mg/L; (C) 8.5 ppm; (D) Cannot be determined.

6. In the BOD_5 determination of sewage, 10 mL of sewage is mixed with enough water to fill a 400 mL bottle. The initial dissolved oxygen is measured to be 6.5 mg/L. However, after 5 days of incubation at 20 °C, the dissolved oxygen concentration decreased by 2.5 mg/L. The BOD_5 of the sewage would be
(A) 100 ppm; (B) 100 mmol/L; (C) 160 mol/L; (D) 160 mmol/L.

7. Which of the following statements regarding water treatment technologies is **NOT** true?
(A) Sedimentation is a separation technology that separates solids from water by using gravity;
(B) A jar test is a lab-scale test that simulates coagulation and flocculation to find the most effective water treatment chemicals;
(C) Ultrafiltration membranes can be used to separate monovalent ions from multivalent ion mixed water;
(D) Ion exchange membranes are a kind of electro-driven membranes.

8. A well-mixed aeration lagoon rectangular has a 1200 m^3 volume, and it receives 300 m^3 of wastewater for treatment every day. To test the BOD_5 of wastewater, the dilution factor for an unseeded mixture of wastewater and pure water is 0.05. The DO of the mixture is initially 10.0 mg/L, and 5 days later, it drops to 5.0 mg/L. What's the efficiency of this lagoon? (The biodegradation rate constants for the wastewater include endogenous decay constant k_d = 0.08 d^{-1}, maximum specific microbial growth rate μ_m = 1.10 d^{-1}, and the half velocity constant K_s = 76.0 mg BOD_5/L).
(A) 67.43%; (B) 74.79%; (C) 77.13%; (D) 84.79%.

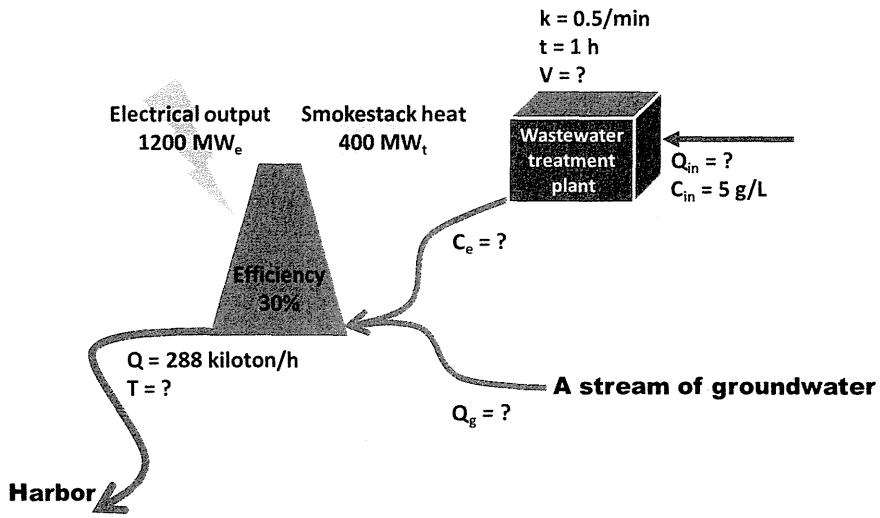
9. Which of the following pollutants is not a major contributor to acid rain formation?
(A) Sulfur dioxide (SO_2);
(B) Nitric oxide (NO);
(C) Carbon monoxide (CO);
(D) Nitric acid (HNO_3).

10. Which of the following is not an ozone destructor?
(A) Halons (CF_3Br; CF_2ClBr);
(B) Chlorofluorocarbons (CFCs);
(C) Hydrochlorofluorocarbons (HCFCs);
(D) Perfluorocarbons (PFCs).

Section B. (Four questions with unequal marks)

<u>You may need to use Appendix A at the end of the exam paper for your calculations.</u>

11. A coal-fired tower generates electrical power with an efficiency of 30% and an output of 1200 MW_e. 400 MW_t of heat is emitted from its smokestack (as shown in below figure). Two streams (in 25 °C) flow through the tower to take away waste heat and are finally discharged into a harbor with a rate of 288 kiloton/h. One is a stream of groundwater. The other is the effluent from a complete-mix wastewater treatment plant. The plant treats 5 g/L of a certain pollutant with a 1st-order reaction rate of 0.5/min and a hydraulic retention time of 1 h. Assuming the streams are in steady state.



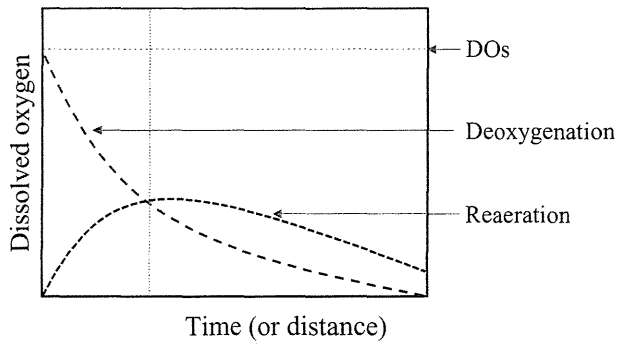
(A) What is the pollutant concentration in the effluent (C_e) of the wastewater treatment plant? **(2 marks)**

(B) **Design** the input flow rate (Q_{in}) of the wastewater treatment plant, based on which to find out its volume (V) and the flow rate of the groundwater stream (Q_g). Briefly state the reasons for the designed value of Q_{in}. **(4 marks)**

(C) What is the temperature of the discharged stream into the harbor? **(4 marks)**

(D) How will the discharged flow from the tower impact the harbor? **(5 marks)**

12. (A) A wastewater has a certain DO concentration and is discharged into a river. The river with a specific flow rate receives this wastewater (Keep the temperature at 20 °C). Assuming complete and instantaneous mixing, please complete the following picture:



Write your answer in the exam answer book.

(i) Please draw out the oxygen sag curve;
(ii) Please point out the minimum DO and oxygen deficit in the oxygen sag curve;
(iii) Please explain the phenomena of deoxygenation and reaeration;
(iv) Please briefly evaluate the wastewater discharge situation in this river. **(8 marks)**

(B) A well-mixed shallow pond has a volume of 300 m^3. Every day, it receives 60 m^3 of wastewater and outflows 60 m^3 of treated effluent. Initially, the microbial concentration in the pond is 0 mg VSS/L, and the substrate concentration is 90 mg BOD_5/L. What's the concentration of VSS leaving the pond? (The biodegradation rate constants for the wastewater include endogenous decay constant k_d = 0.06 d^{-1}, maximum specific microbial growth rate μ_m = 2 d^{-1}, maximum specific substrate utilization rate, k = 6 mg VSS·d, and the half velocity constant K_s = 60.0 mg BOD_5/L).

(6 marks)

(C) Membrane bioreactor (MBR) technology is advancing to improve sludge treatment and resource recovery from complex waste streams like those from hospitals. Based on their configuration, how would you classify MBRs? Briefly explain the primary categories and the key reason for this classification (you may use a diagram). Furthermore, what are the main advantages and disadvantages of using MBR technology for sludge treatment? **(6 marks)**

13. The finite lifespan of electric vehicle batteries (EVBs, 3-10 years) results in a growing annual accumulation of spent EVBs, exceeding 2 million tons annually, and the amount is projected to rise as the demand continues to increase. In response to the directives of sustainable energy, economy, and environment in Hong Kong and worldwide, the recovery of ion resources (e.g., Li^+, Na^+, Ni^{2+}, and PO_4^{3-}) from EVBs has become critical. Membrane technology, with green, efficient, less energy-intensive, and scalable separation processes, has been explored in harvesting ion resources from streams.

(A) Briefly classify the polymeric membrane categories based on (i) membrane pore size, (ii) driving force, and (iii) other classifications, respectively. **(6 marks)**

(B) Please provide an integrated process (including different kinds of membrane technologies) for spent EVB treatment and valuable ions (i.e., Li^+, Ni^{2+}, and PO_4^{3-}) recovery. **(< 400 words including explanations of your design purposes, up to 1 Figure is allowed. Note: $2OH^- + CO_2 + 2Li^+ \rightarrow Li_2CO_3\downarrow + H_2O$).** **(14 marks)**

14. Compared to mobile sources, power plants (especially coal-fired power plants) are major contributors to SOx and emit significant amounts of NOx and particulate matter. Various studies have examined methods to control and reduce these emissions.

(A) List and briefly describe the three main types of air pollution control measures used in power plants. **(3 marks)**

(B) What does SCR stand for, and how does this technology reduce NOx emissions from power plants? Explain the process involved. **(4 marks)**

(C) A temperature inversion can lead to stable atmospheric conditions and poor dispersion of air pollutants. a) Explain what a temperature inversion is. b) Define the adiabatic lapse rate. **(4 marks)**

(D) Describe the conditions that lead to atmospheric instability (unstable). Why does an air parcel that moves adiabatically (without heat exchange) continue to rise or sink under unstable conditions? **(4 marks)**

Appendix A. Information for Reference

I. Atomic weights

H 1; C 12; N 14; O 16; F 19; Na 23; Mg 24.3; Al 27; P 31; S 32; Cl 35.5; K39; Ca 40; Cr 52; Mn 55; Fe 55.8; Ni 58.7; Cu 63.5; Zn 65.4; I 127.

II. Properties of water

Density = 1000 kg/m^3
Viscosity (15 °C) = 1.1404 mPa·s
Specific heat = 4.18 kJ/(kg·°C)
Heat of vaporization (15 °C) = 2465 kJ/kg
Heat of fusion = 333 kJ/kg

III. Other constants

Ideal gas law constant = 0.082056 L·atm/(K·mol)
Henry's Law constant for CO_2: 0.0334 mol/(L·atm)
Acidity constants for H_2CO_3: $pK_1 = 6.3$; $pK_2 = 10.3$
Solubility of $CaCO_3(s)$: $pK_{so} = 8.3$
Henry's Law constant for O_2 (25 °C) = 0.00126 mol/(L·atm)
Dissociation constant of water (25 °C) $K_w = 1\times10^{-14}$
Equivalent Weight (EW) for Ca^{2+} = 20 mg/meq
EW for Mg^{2+} = 12.2 mg/meq
EW for $CaCO_3$ = 50 mg/meq

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